

(48) Иисувовать на скусимлост
и равкоммерную скусимлост иитираи
Дириние.

$$I(\alpha) = \int_0^{+\infty} \frac{\sin \alpha x}{x} dx - \text{Иитираи Дириние}$$

1) $\alpha > 0$

$$\varphi(\alpha, \beta) = \int_0^{+\infty} e^{-\beta x} \frac{\sin \alpha x}{x} dx$$

$\forall \beta > \varepsilon > 0 \quad \forall \alpha = 0 \quad \varphi(\alpha, \beta)$ отограсс

$$\varphi(\alpha, \beta) \sim \frac{1}{x} e^{-\beta x} \downarrow \rightarrow 0$$

пр. Дириние

$$\left| \int_0^x \sin \alpha t \right| \leq \frac{2}{\alpha}$$

2) $\alpha = 0, I(0) = 0$

3) $K(\alpha, \beta) = \int_0^{+\infty} \left(e^{-\beta x} \frac{\sin \alpha x}{x} \right)'_{\alpha} dx =$

$$= \int_0^{+\infty} e^{-\beta x} \cos \alpha x dx - \text{свойства}$$

равномерно при $\alpha \geq 0$

$$\Rightarrow \varphi'_{\alpha}(\alpha, \beta) = r(\alpha, \beta) =$$

$$= \int_0^{+\infty} e^{-\beta x} \cos \alpha x dx = \frac{-\beta \cos \alpha x + \alpha \sin \alpha x}{\alpha^2 + \beta^2} e^{-\beta x} \Big|_0^{+\infty} =$$

$$= 0 - \left(-\frac{\beta}{\alpha^2 + \beta^2} \right) = \frac{\beta}{\alpha^2 + \beta^2}$$

$$\int_0^{\alpha} \varphi'_{\alpha}(\alpha, \beta) d\alpha = \int_0^{\alpha} \frac{\beta}{t^2 + \beta^2} dt \Rightarrow$$

$$\Rightarrow \varphi(\alpha, \beta) - \underbrace{\varphi(0, \beta)}_{=0} = \arctg \frac{\alpha}{\beta}$$

$$\varphi(\alpha, \beta) = \arctg \frac{\alpha}{\beta}$$

$\varphi(\alpha, \beta)$ равномерно сходящаяся по β

$$F = \left| \int_0^x \sin \alpha x dx \right| \leq \frac{2}{\alpha}$$

$$g = \frac{e^{-\beta x}}{x} \downarrow$$

$$q \Rightarrow 0$$

$$\frac{e^{-\beta x} \cos \delta x}{x} \text{ непрекъснато на } [0, \beta] \times [0, 1]$$

$$q^2(\alpha, \beta) \in C^0 [0, 1]$$

$$\lim_{\beta \rightarrow +0} q^2(\alpha, \beta) = I(\alpha)$$

$$\lim_{\beta \rightarrow +0} \arctan \frac{\alpha}{\beta} = \frac{\pi}{2} = I(\alpha)$$

$$\text{т. нар. } I(\alpha) = -I(-\alpha)$$

$$I(\alpha) = \begin{cases} \frac{\pi}{2}, & \alpha > 0 \\ 0, & \alpha = 0 \\ -\frac{\pi}{2}, & \alpha < 0 \end{cases} = \frac{\pi}{2} \operatorname{sgn} \alpha$$



